🏥 Royal Children's Hospital Melbourne: Gen Agent Care Portal

Executive & Technical Project Narrative

This document provides a comprehensive narrative of the RCH Melbourne Care Portal and its Generative AI Agent Integration, outlining the clinical use cases, user personas, conversational workflows, and strategic outcomes for sharing with your broader team and stakeholders.

🌟 Executive Summary

The RCH Care Portal is a state-of-the-art digital paediatric support application integrated with Google Cloud's Vertex AI Customer Engagement Suite (CES).

Moving a child from an acute hospital ward to home care is one of the most stressful transitions a parent can experience. This portal solves that challenge. By combining a premium paediatric portal interface with a cooperative multi-agent conversational engine, we provide parents with 24/7 clinical recovery guidance, real-time safety triage, and seamless home care scheduling - all backed by official hospital guidelines and synchronised directly with the hospital's Electronic Medical Records (EMR).

🎯 The Core Use Case: High-Vulnerability Paediatric Discharge Support

Our target use case is Post-Discharge Recovery Management for Paediatric Asthma and RSV (Respiratory Syncytial Virus).

The Problem

Asthma and RSV are the leading causes of paediatric hospital admissions in Australia. Post-discharge care requires strict medication schedules (spacer mask use) and vigilant symptom monitoring. Stressed parents often struggle to remember dense paper instructions, leading to:

1. Unnecessary ER visits: Parents panicking over minor, manageable symptoms.
2. Delayed care (Clinical Risk): Parents failing to recognize silent "red-flag" symptoms of severe respiratory distress.
3. Administrative overhead: High staff load spent manually scheduling follow-up nurse visits and managing funding packages (NDIS/Medicare).

The Solution

The Gen Agent Care Portal serves as a 24/7 clinical companion that dynamically answers recovery questions using grounded official guidelines, schedules specialised home care nurse visits, and instantly triages life-threatening symptoms.

👥 Key Personas

1. The Parent: Patrick Marlow

- Background: Stressed, exhausted, and balancing work while caring for his recovering child at home.
- Needs: Fast, authoritative, and simple answers. He wants to know exactly how to manage Leo's inhaler, what symptoms are normal, and how to schedule a nurse to visit their home.

2. The Patient: Leo (5 Years Old)

- Diagnosis: Recovering from acute bronchiolitis (RSV) and mild asthma.
- Risk Profile: High vulnerability to respiratory relapses, requiring structured spacer medication and home observation.

3. The Discharge Assistant Agent (AI)

- Role: Patrick's primary clinical companion.
- Capabilities: Highly empathetic, answers recovery questions, searches grounded RCH practice guidelines, and recommends support materials.

4. The Home Care Specialist Agent (AI)

- Role: The administrative and billing orchestrator.
- Capabilities: Modelling home care costs, models government/hospital funding subsidies (NDIS), retrieves open calendar slots, and books appointments.

🛠️ The Demo Experience: Three Key Pillars

Our live demonstration is structured around three high-impact pillars that showcase the agent's intelligence, safety, and integration capabilities:

Demo Script: User Journey

Parent: "Hi"

Agent: "Hello Patrick! Warm welcome back to the Royal Children's Hospital Melbourne. I see we've already got some home care items in your RCH ongoing support plan for Leo. How can I help you and Leo today with Leo's Asthma and RSV recovery today?"

Parent: "I have received this patient discharge form and I need to understand what specific things I need to know about Leo."

Agent: "To give you the most accurate details, please upload a photo of his discharge sheet."

(User uploads the form or shows on video) Sample is attached below.

Agent: [The agent reads the diagnosis of RSV and Asthma and provides tailored suggestions based on that information.]

Example - For Leo's recovery from Asthma and RSV, the most important next steps typically involve:

1. **Monitoring symptoms:** Watch for any increased breathing effort, wheezing, or persistent cough.
2. **Medication management:** Giving any prescribed reliever (like Ventolin) or preventer medications as directed.
3. **Rest and hydration:** Ensuring he gets plenty of rest and sips fluids regularly.

I would love to recommend some specialized pediatric home care resources, such as a **Daily Symptom & Inhaler Tracker Guide** and an **Asthma Spacer Kit**, to help you manage his recovery at home.
What do you think of these ongoing care resources?

Parent: Help me book a nurse visit and add Asthma Spacer Kit to Leo's care plan

Agent : I can certainly help you coordinate a specialized pediatric nurse home visit to monitor Leo's asthma and RSV recovery. I've also noted your request to add the Asthma Spacer Kit to his care plan.
For the nurse visit, we have the following times available tomorrow, July 6th:

- 8:00 AM - 12:00 PM
- 1:00 PM - 5:00 PM

Would either of those windows work for you? Once we've settled on a time, I can also provide a cost estimate and discuss any available funding subsidies.

<Conversation goes on>

THE ROYAL CHILDREN'S HOSPITAL MELBOURNE

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INPATIENT DISCHARGE SUMMARY

Disclaimer: *This is a fictitious sample created for illustrative and educational purposes only. It does not represent a real patient and should not be used as clinical medical advice.*

PATIENT DETAILS

Patient Name: Leo Sample
UR Number: XXXXXXX
Date of Birth: DD/MM/YYYY (Age: 4 years)
Gender: Male
Address: [Patient Address]
General Practitioner: Dr. [GP Name], [Clinic Name]

ADMISSION DETAILS

Date of Admission: 12-July-202X
Date of Discharge: 15-July-202X
Admitting Unit: General Medicine
Consultant: Dr. [Consultant Name]

DIAGNOSES

1. **Principal Diagnosis:** Acute moderate-to-severe asthma exacerbation.
2. **Secondary Diagnosis:** Respiratory Syncytial Virus (RSV) infection.

PRESENTING PROBLEM

Leo is a 4-year-old boy with a known history of viral-induced wheeze/asthma who presented to the Emergency Department with a 3-day history of coryzal symptoms, low-grade fevers, and a 24-hour history of increasing work of breathing, persistent cough, and audible wheeze.

CLINICAL COURSE AND MANAGEMENT

Emergency Department:
Upon arrival in the ED, Leo was tachycardic and tachypnoeic with moderate subcostal and intercostal recessions. Auscultation revealed widespread expiratory wheeze and decreased air entry bilaterally. Oxygen saturations were 90% on room air. He was immediately commenced on a severe asthma pathway. He received bursts of inhaled Salbutamol (Ventolin) and Ipratropium bromide (Atrovent) via a spacer, alongside a stat dose of oral Prednisolone. Due to persistent work of breathing, he was briefly placed on high-flow nasal cannula (HFNC) oxygen support and admitted to the General Medicine ward.

Pillar 1: Authoritative Guidance (Real-Time Clinical Grounding)

- The Interaction: Patrick asks, "How many puffs of Ventolin should I give in an asthma flare-up?"
- The Technology: The agent dynamically searches the official RCH Clinical Practice Guidelines (CPGs) database on Google Cloud.
- The Value: Instead of hallucinating, the agent returns 100% verified paediatric protocols directly from the hospital's database, citing correct inhaler techniques.

Pillar 2: Clinical Safety (Red-Flag Symptoms Guardrails)

- The Interaction: Patrick inputs, "Leo's chest is sucking in deeply and he is grunting."
- The Technology: Natural Language Processing (NLP) classifiers immediately detect clinical "red-flag" indicators of severe respiratory distress (retractions and grunting).
- The Value: The agent instantly halts all standard operations (scheduling, educational chats) and displays a prominent emergency card directing Patrick to call 000 or present to the nearest Emergency Department, acting as a critical safety net.

Pillar 3: Administrative Streamlining (Multi-Agent & EMR Sync)

- The Interaction: Patrick requests a home nurse visit. The Discharge Agent seamlessly hands off to the Home Care Specialist. The agent displays available times, calculates NDIS funding, and registers the booking.
- The Technology: Dynamic multi-agent routing, live mathematical billing calculations, and direct synchronisation with the hospital EMR database.
- The Value: Eliminates the administrative friction of follow-up care, reducing nurse scheduling desk calls to zero.

📈 Strategic Business & Clinical Outcomes

By implementing the Gen Agent Care Portal, the hospital achieves major clinical and organisational milestones:

Metric / Area	Traditional Process	With Gen Agent Portal

Patient Safety & Care	Parents rely on memory or unverified internet searches during panic.	Grounded, official RCH medical guidelines are available 24/7 with instant emergency safety intercepts.
Hospital Readmission Rates	High relapse rates due to incorrect inhaler technique or missed follow-ups.	Active monitoring and friction-free nurse scheduling keep children recovering safely at home.
Administrative Friction	Staff spend hours on the phone managing calendars, billing, and NDIS paperwork.	Automation of cost estimations, funding registrations, and calendar lookups.
EMR Logging Accuracy	Manual charting and administrative notes are logged with delays.	Real-time background sync directly into the Hospital EMR database.

NOTE

Summary for the Team: This demo proves that Gen AI in healthcare can be safe, integrated, and highly interactive. It respects clinical boundaries, enforces patient safety, and streamlines administrative work—directly improving both parent reassurance and clinical outcomes.